

A BOUNDING-NUMBER VERSION OF THE DENSITY-KILLING STEP

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SOURCE AND TARGET

Drygier and Plebanek call a compactification $\gamma\omega$ *smooth* when the natural copy of c_0 is complemented in $C(\gamma\omega)$ [1]. In Boolean-algebra language, if $\text{Fin} \subseteq \mathcal{A} \subseteq P(\omega)$, the compactification $K_{\mathcal{A}}$ is smooth if and only if there is a bounded sequence (ν_n) of finitely additive measures on $\mathcal{A}$, vanishing on finite sets, such that

$$\nu_n(A) - \delta_n(A) \longrightarrow 0 \quad (A \in \mathcal{A}).$$

The source proves under CH that there is a smooth compactification of ω with nonseparable remainder, and explicitly notes that it is not known whether the CH assumption can be removed; the authors suggest that $\mathfrak{b} = \mathfrak{c}$ may suffice.

The present note proves the local strengthening behind that suggestion. The countable Boolean algebra in Lemma 6.3 of [1] can be replaced by an arbitrary Boolean algebra of cardinality $< \mathfrak{b}$. This is not by itself the full $\mathfrak{b} = \mathfrak{c}$ theorem, because the global bookkeeping of all possible countable dense extension candidates remains a separate issue.

IDEA OF THE PROOF

In the source proof, countability of $\mathcal{B}$ is used to enumerate $\mathcal{B}$ and force every old algebra element contained in the new set X to be contained in a finite initial part of X . The companion paper [2] proves precisely the replacement for this enumeration: for an algebra of size $< \mathfrak{b}$ and a countable decreasing s -family $(S_j(k))_k$, one can choose a single function $g \in \omega^\omega$ so that every $B \in \mathcal{B}$ with $B \subseteq \bigcup_j S_j(g(j))$ is already contained in a finite subunion.

We build such s -families inside the prescribed dense ultrafilters p_j . The sets $S_j(k)$ are chosen small for finitely many measures, absorb the finite exceptional set where pointwise convergence has not yet made them small, and avoid longer and longer initial segments of ω . Then

$$X = \bigcup_j S_j(g(j))$$

belongs to every extension of every p_j , while $\omega \setminus X$ remains infinite. Thus those extensions are not dense. The same finite-subunion property gives the

required inner-measure estimate $(\nu_n)_*(X) \rightarrow 0$ along $n \notin X$, while density of the p_j 's gives $\nu_n^*(X) = 1$. Los–Marczewski extensions then preserve $\nu_n - \delta_n \rightarrow 0$.

DEFINITIONS

Let $\mathfrak{b}$ denote the bounding number. If $\text{Fin} \subseteq \mathcal{B} \subseteq P(\omega)$, write $K_{\mathcal{B}}$ for the Stone compactification of ω and $K_{\mathcal{B}}^*$ for its remainder. For a finitely additive probability ν on $\mathcal{B}$ and $Z \subseteq \omega$, define

$$\nu_*(Z) = \sup\{\nu(B) : B \in \mathcal{B}, B \subseteq Z\}, \quad \nu^*(Z) = \inf\{\nu(B) : B \in \mathcal{B}, Z \subseteq B\}.$$

All measures below are nonnegative finitely additive probabilities.

Lemma 1 (finite-cover lemma). *Let $\text{Fin} \subseteq \mathcal{B} \subseteq P(\omega)$ have $|\mathcal{B}| < \mathfrak{b}$. For each $j \in \omega$, let $(S_j(k))_{k \in \omega}$ be a decreasing sequence in $\mathcal{B}$ with $\bigcap_k S_j(k) = \emptyset$. Then there is $g_0 \in \omega^\omega$ such that whenever $g \geq g_0$ and*

$$X_g = \bigcup_{j \in \omega} S_j(g(j)),$$

every $B \in \mathcal{B}$ satisfying $B \subseteq X_g$ is contained in a finite subunion $\bigcup_{j \leq N} S_j(g(j))$.

Proof. This is the domination lemma used in [2]; we recall the argument to make clear that only $|\mathcal{B}| < \mathfrak{b}$ and the decreasing-to-empty property are used.

For a finite set $m = \{0, \dots, m-1\}$ and $\varphi \in \omega^m$, write

$$X_\varphi = \bigcup_{j < m} S_j(\varphi(j)).$$

For fixed $B \in \mathcal{B}$, the set of $\varphi \in \omega^m$ for which $B \setminus X_\varphi$ is infinite has finitely many minimal elements under the coordinatewise order; this is the finite-dimensional combinatorial lemma from [2, Lemma 4.1], proved by induction on m . Consequently, for every m there is a finite set $I_m^B \subseteq \omega$ such that whenever $B \setminus X_\varphi$ is infinite, it meets I_m^B .

Choose $h_B \in \omega^\omega$ so that

$$S_j(h_B(j)) \cap \bigcup_{m \leq j} I_m^B = \emptyset \quad (j \in \omega).$$

This is possible because each $S_j(k)$ decreases to the empty set. Since $|\mathcal{B}| < \mathfrak{b}$, choose g_0 eventually dominating all h_B , $B \in \mathcal{B}$.

Let $g \geq g_0$, $B \in \mathcal{B}$, and $B \subseteq X_g$. Pick N such that $h_B(j) \leq g(j)$ for $j \geq N$. If $B \setminus X_{g|N}$ were infinite, then it would meet I_N^B . But every tail set $S_j(g(j))$, $j \geq N$, is disjoint from I_N^B , contradicting $B \subseteq X_g$. Hence $B \setminus X_{g|N}$ is finite. Each of its finitely many points lies in some tail $S_j(g(j))$, so increasing N if necessary gives $B \subseteq \bigcup_{j \leq N} S_j(g(j))$. $\square$

THE LOCAL THEOREM

Theorem 2. *Let $\text{Fin} \subseteq \mathcal{B} \subseteq P(\omega)$ be a Boolean algebra with $|\mathcal{B}| < \mathfrak{b}$. Let $(p_j)_{j \in \omega}$ be a countable dense set in $K_{\mathcal{B}}^*$. Suppose that $(\nu_n)_{n \in \omega}$ is a sequence of nonatomic probabilities on $\mathcal{B}$ such that*

$$\nu_n|_{\text{Fin}} = 0 \quad \text{and} \quad \nu_n - \delta_n \rightarrow 0 \quad \text{pointwise on } \mathcal{B}.$$

Then there is an infinite set $X \subseteq \omega$ such that:

- (i) *for every choice of extensions $\tilde{p}_j \in \text{ult}(\mathcal{B}[X])$ of p_j , the family $\{\tilde{p}_j : j \in \omega\}$ is not dense in $K_{\mathcal{B}[X]}^*$;*
- (ii) *every ν_n has a nonatomic probability extension $\tilde{\nu}_n$ to $\mathcal{B}[X]$, and these extensions can be chosen so that $\tilde{\nu}_n - \delta_n \rightarrow 0$ pointwise on $\mathcal{B}[X]$.*

Proof. Put $\eta_j = 2^{-(j+3)}$. For each j we first construct a decreasing sequence $S_j(k) \in p_j$, $k \in \omega$, such that

$$\bigcap_k S_j(k) = \emptyset$$

and, for every k ,

$$(1) \quad n \notin S_j(k) \implies \nu_n(S_j(k)) < \eta_j.$$

We also arrange $\nu_i(S_j(k)) < \eta_j$ for all $i \leq k$.

Assume $S_j(k-1)$ has been chosen, with $S_j(-1) = \omega$. Since the finite sum $\nu_0 + \dots + \nu_k$ is nonatomic and p_j is nonprincipal, there is

$$C \in p_j, \quad C \subseteq S_j(k-1) \setminus \{0, \dots, k-1\},$$

such that $\nu_i(C) < \eta_j$ for $i \leq k$. By pointwise convergence on $\mathcal{B}$, the set

$$F = \{n \notin C : \nu_n(C) \geq \eta_j\}$$

is finite. For $k > 0$, the inductive hypothesis applied to $S_j(k-1)$ gives $F \subseteq S_j(k-1)$. Also $F \cap \{0, \dots, k-1\} = \emptyset$. Define

$$S_j(k) = C \cup F.$$

Finite sets have ν_n -measure zero, so the smallness inequalities are unchanged. The sequence is decreasing and avoids every fixed integer eventually, hence has empty intersection.

We now use Lemma 1. There is some $g \in \omega^\omega$ such that, for

$$X = \bigcup_{j \in \omega} S_j(g(j)),$$

every $B \in \mathcal{B}$ satisfying $B \subseteq X$ is contained in a finite subunion $\bigcup_{j \leq N} S_j(g(j))$.

The set $\omega \setminus X$ is infinite. Indeed, if $\omega \subseteq X$, the finite-subunion property would give $\omega \subseteq \bigcup_{j \leq N} S_j(g(j))$ for some N , and therefore

$$1 = \nu_0(\omega) \leq \sum_{j \leq N} \nu_0(S_j(g(j))) \leq \sum_{j \leq N} \eta_j < \frac{1}{2},$$

a contradiction.

Since $S_j(g(j)) \in p_j$ and $S_j(g(j)) \subseteq X$, every extension of p_j to $\mathcal{B}[X]$ contains X . The nonempty clopen set $\widehat{\omega \setminus X} \cap K_{\mathcal{B}[X]}^*$ is therefore disjoint from all extensions of the p_j 's. This proves (i).

We next estimate the inner and outer measures of X . Fix n . If $B \in \mathcal{B}$ and $\nu_n(B) > 0$, then B is infinite. Since (p_j) is dense in $K_{\mathcal{B}}^*$, some p_j contains B . Hence

$$B \cap S_j(g(j)) \in p_j$$

is infinite, so $B \cap X \neq \emptyset$. Thus no positive-measure member of $\mathcal{B}$ is disjoint from X , and consequently

$$(2) \quad \nu_n^*(X) = 1.$$

For the inner measure, let $n \notin X$, and let $B \in \mathcal{B}$ with $B \subseteq X$. By the finite-subunion property,

$$B \subseteq \bigcup_{j \leq N} S_j(g(j))$$

for some N . Given $K \leq N$, split this finite union into the first K pieces and the tail. Since $n \notin X$, condition (1) gives

$$\nu_n \left(\bigcup_{K \leq j \leq N} S_j(g(j)) \right) \leq \sum_{j \geq K} \eta_j.$$

For each fixed $j < K$, pointwise convergence on $S_j(g(j))$ gives $\nu_n(S_j(g(j))) \rightarrow 0$ along $n \notin X$, because then $\delta_n(S_j(g(j))) = 0$. Therefore

$$\limsup_{n \notin X} (\nu_n)_*(X) \leq \sum_{j \geq K} \eta_j.$$

Letting $K \rightarrow \infty$, we obtain

$$(3) \quad \lim_{n \notin X} (\nu_n)_*(X) = 0.$$

Define $t_n = 1$ for $n \in X$, and

$$t_n = (\nu_n)_*(X)$$

for $n \notin X$. By (2), $t_n \in [(\nu_n)_*(X), \nu_n^*(X)]$. The Los–Marczewski extension theorem, used in the source as Theorem 3.2, gives a probability extension $\tilde{\nu}_n$ of ν_n to $\mathcal{B}[X]$ with $\tilde{\nu}_n(X) = t_n$. By construction and (3),

$$\tilde{\nu}_n(X) - \delta_n(X) \rightarrow 0.$$

The source convergence lemma [1, Lemma 3.3] then implies

$$\tilde{\nu}_n - \delta_n \rightarrow 0 \quad \text{pointwise on } \mathcal{B}[X].$$

Finally, the extensions may be chosen nonatomic. If $q \in \text{ult}(\mathcal{B}[X])$ and $p = q|_{\mathcal{B}}$, nonatomicity of ν_n gives $B \in p$ with $\nu_n(B) < \varepsilon$. One of $B \cap X$ and $B \setminus X$ belongs to q , and its $\tilde{\nu}_n$ -measure is at most $\nu_n(B)$. Hence $\inf\{\tilde{\nu}_n(C) : C \in q\} = 0$, as required. $\square$

CONSEQUENCES AND LIMITATION

Theorem 2 is exactly the source paper’s density-destruction step with “countable” replaced by “of cardinality $< \mathfrak{b}$ ”. It therefore removes the local obstacle to running the large smooth compactification construction past countable stages.

It does not by itself prove the full $\mathfrak{b} = \mathfrak{c}$ strengthening of Theorem 6.1 of [1]. In a long construction one must also ensure that every possible countable dense family of ultrafilter extensions is presented at a stage where the current algebra still has size $< \mathfrak{b}$. That global bookkeeping problem is separate from the local lemma proved here.

NOVELTY CHECK

The run indexes contained no previous packet or attempt for arXiv:1601.03770. A bounded web search for the exact title, “smooth compactification”, “non-separable remainder”, and “ $\mathfrak{b} = \mathfrak{c}$ ” found the source paper and the companion arXiv:1501.06972 result on nonseparable remainders supporting a measure, but no later smoothness theorem. The present statement is not claimed as a full literature resolution of the source question.

HUMAN REVIEW RECOMMENDATION

Review Theorem 2 as a technical partial result. The delicate points are the recursive construction of $S_j(k)$, the use of the $< \mathfrak{b}$ finite-subunion lemma, and the passage from the inner/outer measure estimates to the Los–Marczewski extensions.

REFERENCES

- [1] P. Drygier and G. Plebanek, *Compactifications of ω and the Banach space c_0* , arXiv:1601.03770, 2016.
- [2] P. Drygier and G. Plebanek, *Nonseparable growth of the integers supporting a measure*, arXiv:1501.06972, 2015.